

Supporting information

The Cyanotype Process and its Potential in Chemistry Education

Alexander Kmet' *, Anna Drozdíková, Soňa Nagyová, Peter Ikhardt

Faculty of Natural Sciences, Comenius University in Bratislava, Department of
Didactics in Science, Psychology and Pedagogy, Ilkovicova 6, Bratislava, Slovakia 84215

* kmet45@uniba.sk

Making cyanotypes – the blue photographs

**A laboratory experiment for elementary school students: ISCED 2 level student
handout**

Course: _____

Name: _____

Instructor: _____

Date: _____

Context:

Some compounds undergo chemical changes when exposed to light, which makes them useful in the process of analogue photography. One of the compounds is ferric ammonium citrate. When mixed with potassium ferricyanide it becomes light-sensitive. After treating paper with this mixture and exposing it to sunlight cyanotype – a blue photograph is produced. One of the first photography pioneers, Anna Atkins used this process to make photograms of plants: a procedure you'll be following during this lesson.

Equipment: 2 small beakers, filtration paper, clip-in frame, brush, scissors

Chemicals: 20 % green ferric ammonium citrate solution, 14 % potassium ferricyanide solution, water

Experimental procedure:

1. Mix 10 ml of **20 % green ferric ammonium citrate** solution with the same amount of **14 % potassium ferricyanide** solution.
2. Spread the mixed solutions onto the paper using a brush.
3. Arrange objects on the paper and put it into the clip frame.
4. Take the frames outside and expose them to sunlight for 4 minutes. The paper should be dark blue after the exposure.
5. Wash the paper under running water until it becomes white.

Questions:

1. What is the difference between artificial light and sunlight? What component of sunlight caused the sensitized paper to undergo the reaction?